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Trustworthy Machine Learning Across the Lifecycle: A Synthesis of Current Methods.

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The interplay between AI and Formal Verification 8th-11th March

INTRODUCTION

Problem

AI systems

Deployed

Critical-domain:
- Autonomous vehicles (ADAS/AV)
- Healthcare
- Financial infrastructures
- Cyber-physical control systems

Failures

Face

Consequences:
- Economic and Financial
- Ethical
- Fatality (loss of human life)

Challenges
- Safety
- Security
- Robustness

INTRODUCTION

Misclassification examples

Fig. 1: Adversarial Attack

Fig. 2: Misinterpretation

INTRODUCTION

Motivation

1. **Fragmentation** and lack of **structured verification** approaches along the ML lifecycle.
2. As in SDLC, AI models should satisfy some requirements to ensuring the **trustworthiness**.
3. **Formal verification** has emerged as promising approach for the V&&V of AI systems

INTRODUCTION

Objectives

- Review existing V&&V applied in AI systems.
- Map them across the ML lifecycle.
- Identify their properties coverages.

METHODOLOGY

Articles extraction

- **Article search and selection related to:** ***AI core concepts***, ***trustworthiness***, ***assurance methods***, and the ***lifecycle***.

- **Filtering process:** No temporal restriction; peer-reviewed articles, conference proceedings, and preprint papers were considered.

- **56** papers were found, and **32** that satisfied **our area of interest** were deeply analyzed for mapping.

METHODOLOGY

Query formulation

Table 1: combination of keywords for articles extraction

AI Core Concepts	Assurance Methods	Trustworthiness	System lifecycle
AI, ML, DL, NN, Data-driven models, Intelligent Systems, dataset, model	*V&&V, Formal Methods, Model Checking, Monitoring, Runtime Verification, Requirement*	*Safety, Security, Robustness, Adversarial attack, Trustworthy, Out-of-Distribution (OOD), poisoning*	*Lifecycle, Pipeline, Framework, Development lifecycle, Deployment, Operational*

Keyword formulation: (*FORMAL VERIFICATION*) **AND** (*AI*)**AND** ((*MODEL*) **OR** (*SYSTEM*))

Formal verification applied to AI system.

METHODOLOGY
Bibliographic sources

Table 2: Digital libraries used for searching related studies

Source
Digital library: IEEE Xplore, ACM Digital Library ★ ★★ ★ ★
Bibliographic database: Compendex, Engineering village, Scopus ★ ★★ ★
Institutional discovery: Sofia (UQAC) ★ ★★ ★
Search Engine: Google Scholar ★ ★★
Preprint: arXiv ★ ★
Social network: ResearchGate ★ ★

METHODOLOGY

ISO/IEC 23053 ML lifecycle

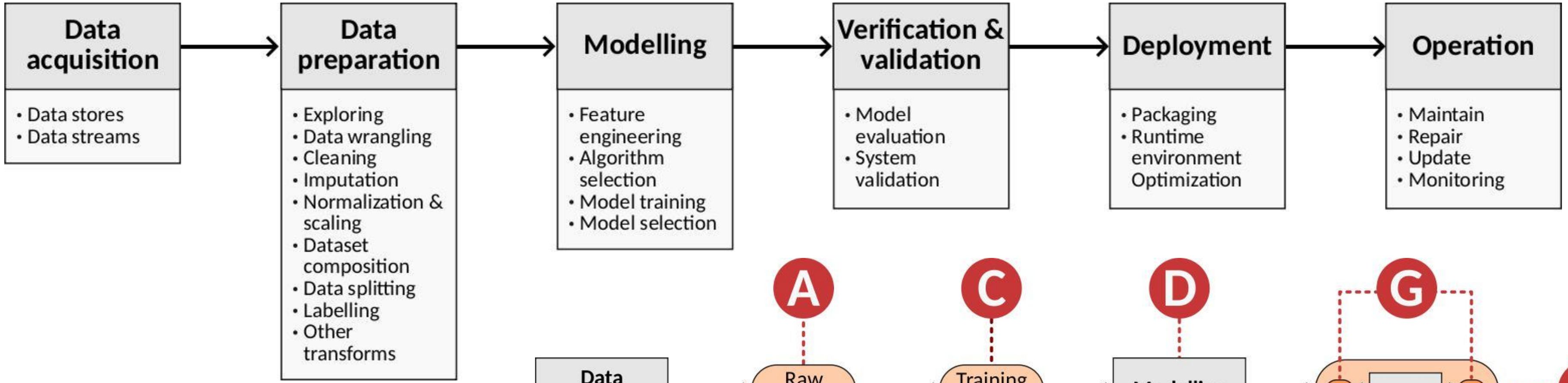

Fig. 1: The ISO/IEC 23053.

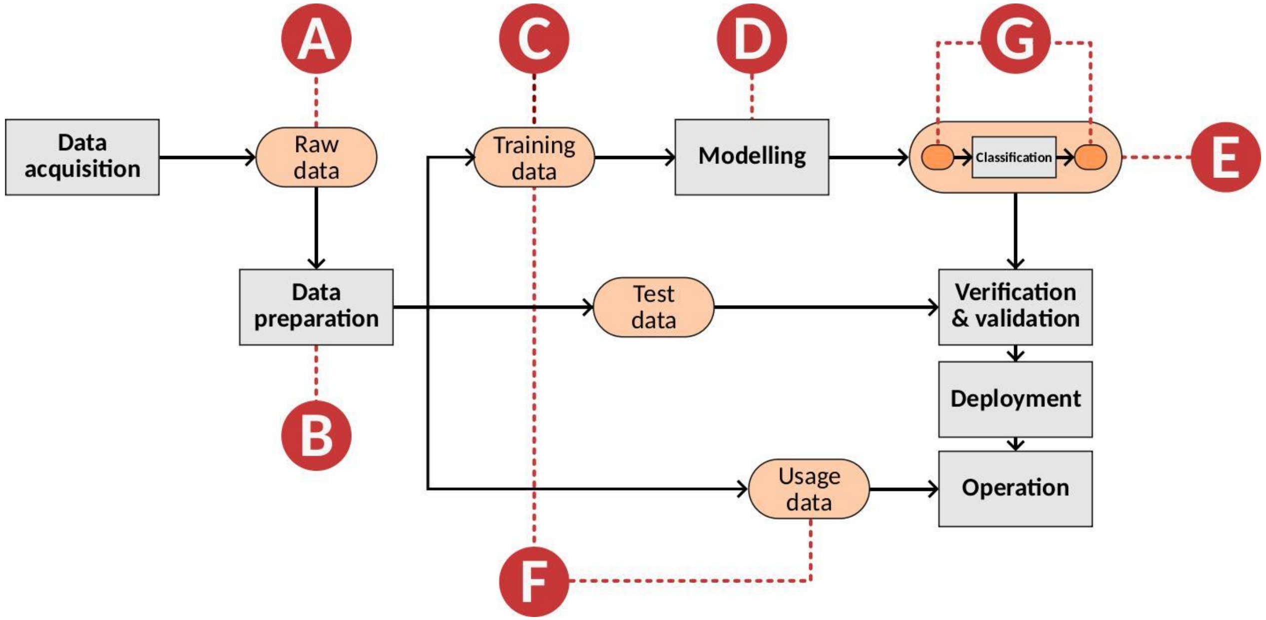

Fig. 2: The expanded ML pipeline.

Results

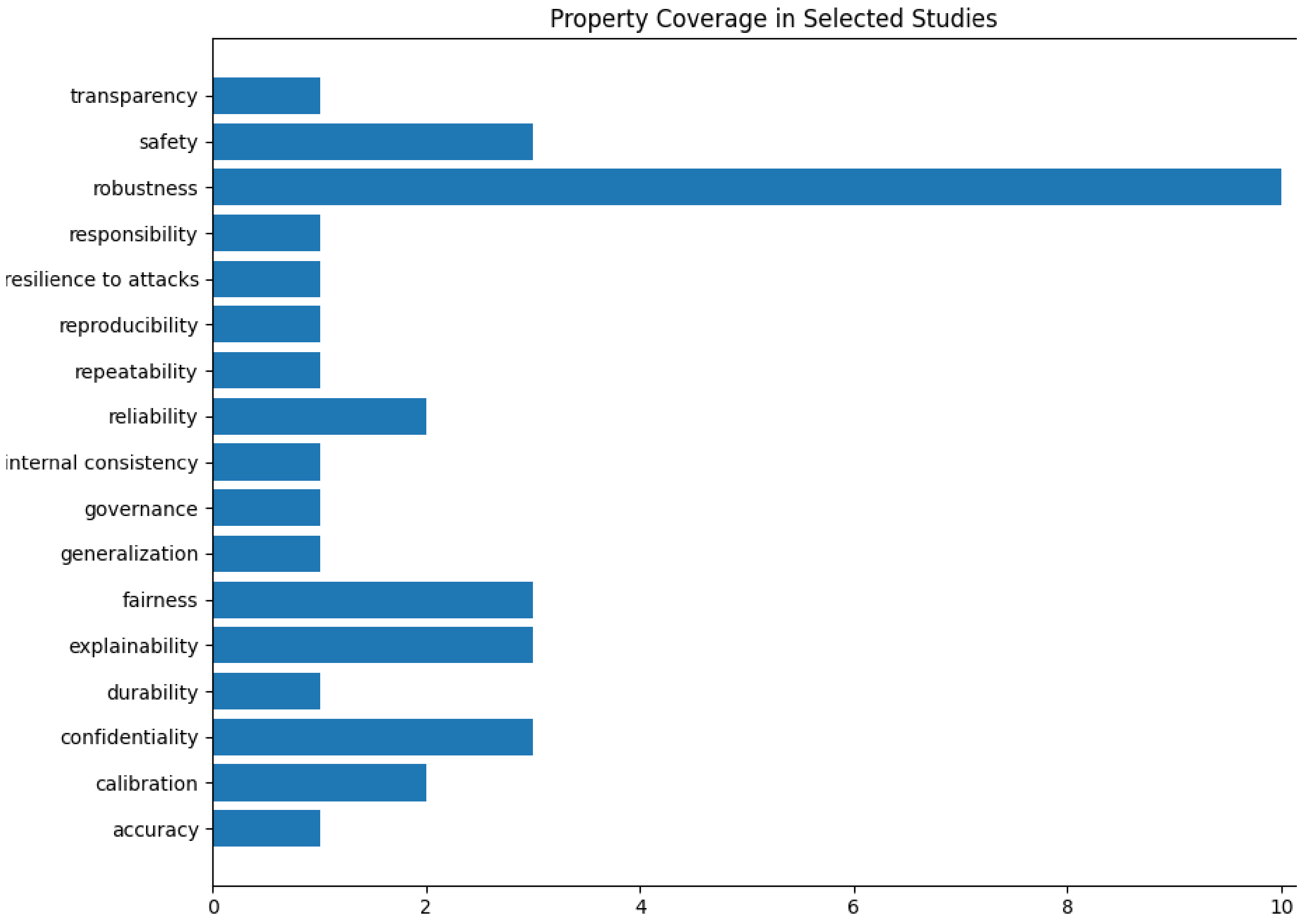

Fig. 4: Property coverage by paper

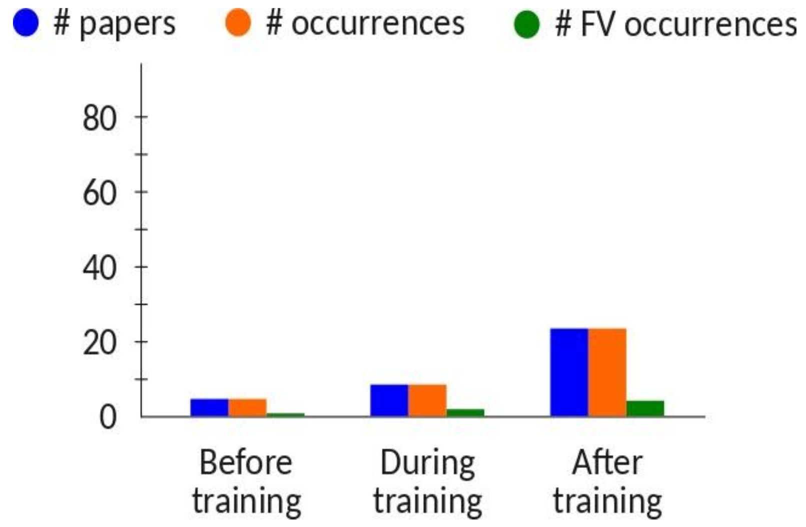

Fig. 3: Distribution of studies across phases.

Segmentation of the extended ML:

- **Before training**: Stage A, B, C
- **Training phase**: Stage D
- **Post training**: Stage E, F, G

Key Findings

- Effort focusing in **post-training**, and concentrated on **robustness** property.
- Few works address **data stage** and **modeling phase.**
- The lifecycle coverage remains limited and limited study on Formal Verification applied to AI.

Conclusion

- ❑ Found **56 papers**, and analyzed deeply the **32**.
- ❑ Mapping verification approaches to the **ML lifecycle**.
- ❑ **Gaps**:
  - Formal verification.
  - Hybrid neuro-symbolic methods.
  - Emerging standard: ISO/IEC 42001, ISO/IEC TR 24029, IEEE 3129-2023, IEEE P7000 series.

To enable earlier detection of errors and reduce the need for later corrective assurance after deployment.

Thank you for your kind attention!